

Significant Figures and Measurement

Answer key with worked solutions

A. Understanding

1. Ruler.

The millimetre marks make 4.3 cm certain, and you can **estimate one more digit** between the marks (4.37 cm). A fourth digit (4.372 cm) would be a guess beyond what the ruler can show.

2. Accuracy and precision.

Accuracy is how close a result is to the true value; **precision** is how close repeated results are to each other. Yes: a balance that reads 0.20 g too high gives results that agree closely (precise) but are all wrong by the same amount (not accurate).

B. Counting significant figures

3. Count.

(a) 2 (b) 4 (c) 4 (d) 5 (e) **ambiguous**: 2, 3 or 4. Write it in scientific notation to show which, e.g. 1.20×10^3 mL for 3.

C. Scientific notation and rounding

4. Scientific notation.

(a) 3.50×10^{-4} m (b) 4.56×10^4 g (c) 2.07×10^{-2} L

5. Round to 3 significant figures.

(a) **2.46 g** (b) **0.00123 m** (c) 1.88×10^4 s (18 800 s would be ambiguous) (d) **10.0 mL** (the first dropped digit is 5, so 9.99 rounds up to 10.0)

D. Calculations

6. Multiply and divide.

(a) $3.25\text{ cm} \times 2.1\text{ cm} = 6.825\text{ cm}^2 \rightarrow \mathbf{6.8\text{ cm}^2}$ (2 s.f.) (b) $100.0\text{ g} \div 3.0\text{ mL} = 33.33\text{ g/mL} \rightarrow \mathbf{33\text{ g/mL}}$ (2 s.f.) (c) $8.315\text{ J/(mol}\cdot\text{K)} \times 298\text{ K} = 2477.87\text{ J/mol} \rightarrow 2.48 \times 10^3\text{ J/mol}$ (3 s.f.; the units of K cancel)

7. Add and subtract.

(a) $12.0\text{ g} + 0.335\text{ g} + 4.5\text{ g} = 16.835\text{ g} \rightarrow \mathbf{16.8\text{ g}}$ (1 d.p.) (b) $6.38\text{ cm} - 2.1\text{ cm} = 4.28\text{ cm} \rightarrow \mathbf{4.3\text{ cm}}$ (1 d.p.)

E. Challenge

8. Aluminium block.

$$V = 2.00\text{ cm} \times 3.00\text{ cm} \times 4.00\text{ cm} = \mathbf{24.0\text{ cm}^3}$$
 (3 s.f.)
$$\rho = \frac{m}{V} = \frac{64.8\text{ g}}{24.0\text{ cm}^3} = \mathbf{2.70\text{ g/cm}^3}$$
 (3 s.f.)

9. Mean.

Sum = $2.52\text{ g} + 2.48\text{ g} + 2.51\text{ g} = 7.51\text{ g}$ (two decimal places). Mean = $7.51\text{ g} \div 3 = 2.5033\text{ g} \rightarrow \mathbf{2.50\text{ g}}$. The 3 is a counted, exact number, so it does not limit the precision.